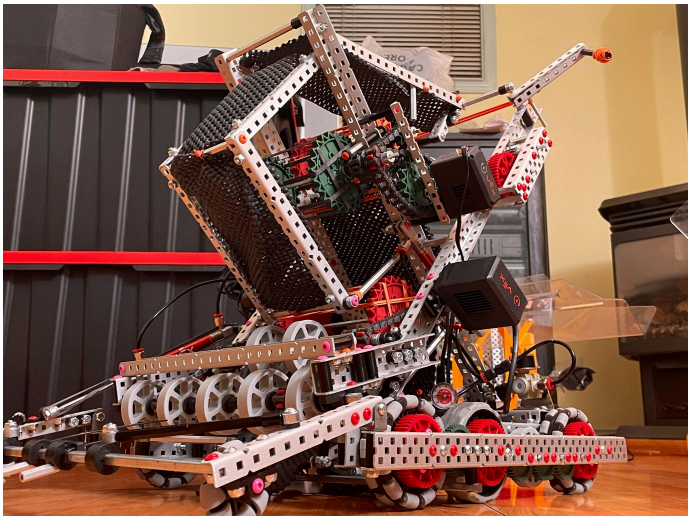
Portfolio

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Resume

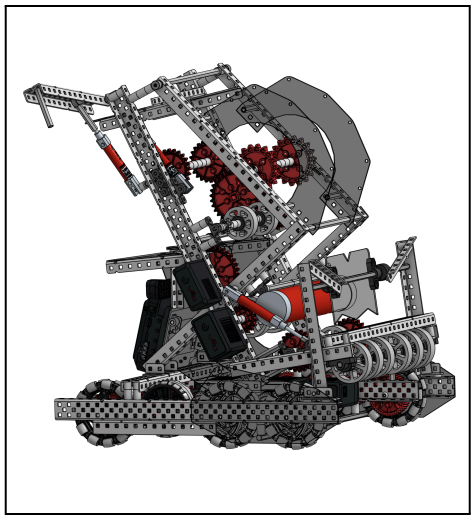
2025-2026 VEX Pushback Robots

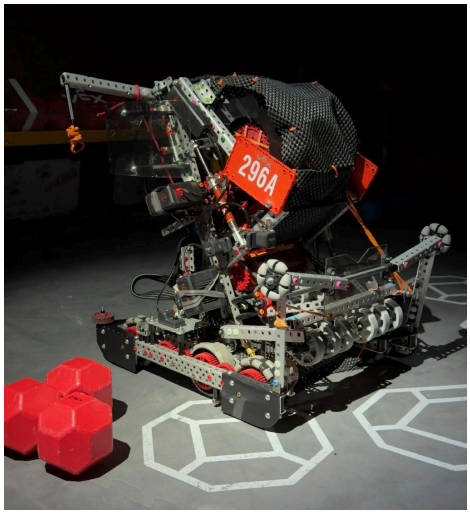


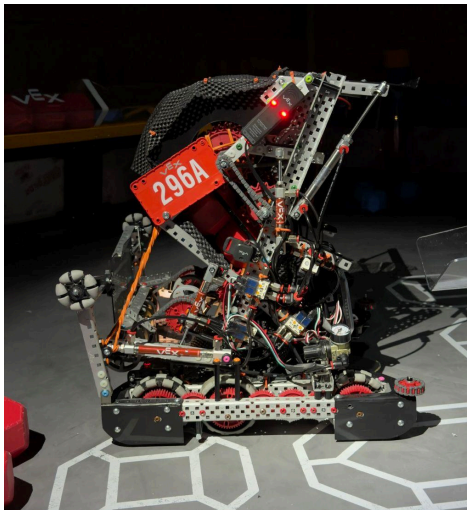


Photos of August – October 2025 Competition Robot

- Devised an S-shaped intake geometry to hold a 14-object capacity within a strict 18 x 18 x 18 inch size limit, tuning roller compression so objects could be collected, tumble freely in the intake, and still outtake quickly without overloading motors
- Designed, laser-cut and heat-bent Delrin sled/aligner parts to improve barrier traversal and alignment during scoring
- Created a sensor-triggered pneumatic lift mechanism to balance a 15-pound robot onto a game object for dual-robot parking, reducing required operating pressure from 60 psi to 30 psi
- Reduced robot weight from 25 lb to <15 lb by simplifying geometries and selecting hardware materials (nylon, aluminum, steel) by application and stress conditions, decreasing drivetrain motor load and increasing acceleration by 30%
- Replaced encoder-only tracking with a Monte Carlo Localization (MCL) system fusing motor encoder, IMU, odometry, and distance-sensor data to reduce positional drift, raising run consistency from 70% to 98%
- Developed a motor current-sensing algorithm to detect intake and outtake events via current spikes, replacing time-based motion chaining, for faster, adaptive autonomous paths, cutting cycle time by 40%
- **Technical documentation available here; code available here**





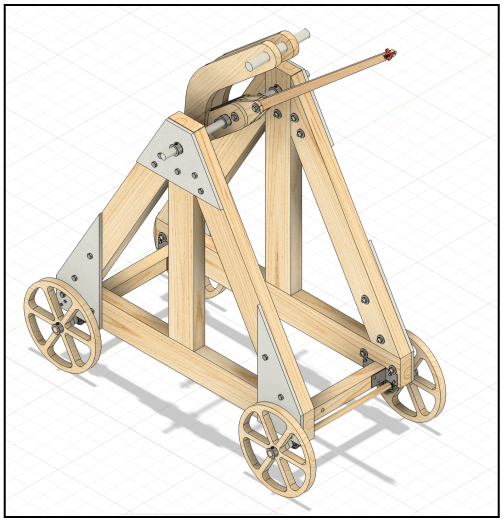


Progress of February 2026 Competition Robot from Initial CAD to Final Product

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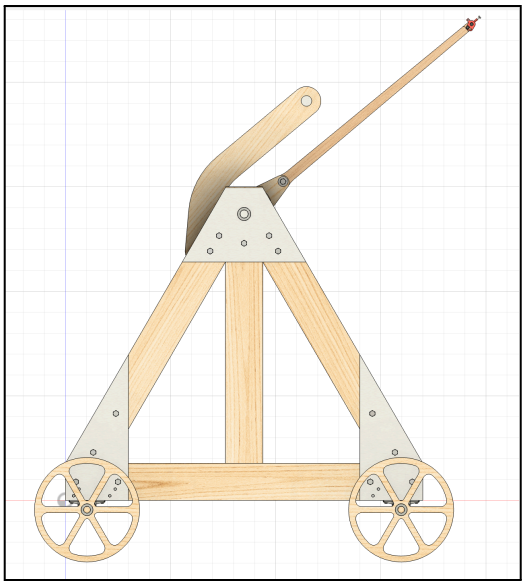
Whipper Trebuchet



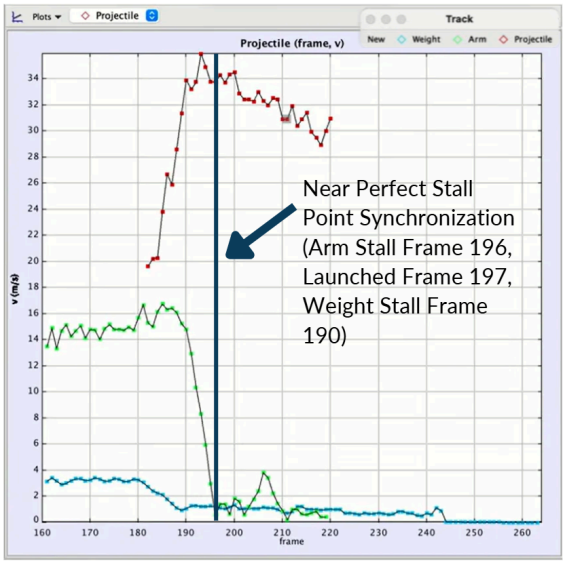


Progress of Trebuchet from CAD to Final Product

- Constructed a frame-mounted wheel system that lets the trebuchet roll forward as the counterweight falls, synchronizing the arm and counterweight stall points to direct more release energy into the throwing arm, increasing energy transfer efficiency from 60.2% to 67.8%
- Implemented the motion-tracking software, Tracker, to plot the position and velocity of the arm, counterweight, and projectile from slow-motion footage frame-by-frame, identifying each component's stall point to precisely tune sling length and release angle for near-perfect synchronization (off by 1.68 ms)
- Diagnosed a consistent ~12-degree deviation in projectile flight path and corrected it by adjusting sling string routing to induce backspin, achieving a near dead-straight trajectory
- Modelled and 3D-printed a custom launch pin enabling quick adjustment of launch angle and sling release timing across different counterweights during testing
- Designed a custom pin-and-washer safety release mechanism for controlled firing of projectiles exceeding 120 km/h



Side View of Trebuchet CAD



Graph of Weight, Trebuchet Arm, & Projectile Velocities